Date .. / /

task 1

東京

1- lw x5, 4(x20)

2- addi x5, x5, 7

3- sw x5, 12(x20)

Task 2

CPU Time = Ic x CPI x CCT

A

Ic = 10^6

CPI = 2

CCT = 25 ps

$T_A = 10^6 \times 2 \times 25$

$T_A = 50,000,000$ ps

convert to sec

$1 ps = 10^{-12} s$

$T_A = 50,000,000 \times 10^{-12}$

$T_A = 0.00005$ s

B

Ic = 10^6

CPI = 1.2

CCT = 500 ps

$T_B = 10^6 \times 1.2 \times 500$

$T_B = 600,000,000$ ps

convert to sec

$T_B = 600,000,000 \times 10^{-12}$

$T_B = 0.0006$ s

$Speedup = \frac{T_B}{T_A} = \frac{0.0006}{0.00005} = 1.2$

Computer A is 1.2 faster than computer B